

THENMOZHI SEKAR

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EDUCATION

Quinnipiac University, Master of Science in Computer Science, GPA – 3.8	Hamden, CT, USA Aug 2024 – May 2026
Madras University, Bachelor of Computer Applications, GPA - 3.7	Kanchipuram, TN, India Aug 2018 – May 2021

EXPERIENCE

Graduate Research Assistant (AI/ML & Software Engineering) Quinnipiac University, Hamden, CT, USA - Designed and implemented machine learning and data engineering pipelines involving preprocessing, feature engineering, LLM-based workflows, and experimental analysis using Python and Java - Conducted AI and Software Engineering research focused on intelligent systems, AI-generated code evaluation, and machine learning-driven applications - Collaborated with faculty advisors on research proposal development, experimentation, technical documentation, and publishable AI/ML research projects	Aug 2024 – Present
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AI-Driven Full-Stack Software Engineer Goaira Technologies Pvt Limited, Chennai, India - Analyzed system performance and user issue data to identify operational patterns, improving reliability across 200+ production devices - Automated monitoring and backend maintenance workflows using scripting and automation tools, reducing manual operational effort by 30% and improving efficiency - Developed data-driven operational workflows and collaborated with cross-functional teams to optimize deployment processes and predictive maintenance initiatives	Jun 2022 – Apr 2024
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Full-Stack Java Software Engineer Ijona Technologies Pvt. Ltd, Bangalore, India - Developed and validated algorithmic models using experimental analysis techniques to improve software performance, scalability, and operational efficiency - Conducted user behavior and system interaction analysis to optimize application design and enhance overall system performance - Improved system reliability through debugging, testing, performance analysis, and continuous optimization across software development workflows	Nov 2021 – May 2022
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PROJECTS

LLM-Based Bug Report Enhancement Pipeline Python, NLP, LLMs - Built an LLM-based NLP pipeline to transform unstructured bug reports into standardized formats, improving report completeness by 72% across 15,000+ GitHub API bug reports - Developed automated preprocessing and structured insight extraction workflows to improve debugging, issue triaging, and reproducibility analysis	Jan 2026 – Present
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AI-Based Code Generation Evaluation Python, Java, Published Research - Conducted empirical evaluation of AI-generated code using 200+ code samples and 1200+ experimental runs across similarity, structural, and logical correctness metrics - Published research at IISec 2026 focused on assessing AI-assisted software engineering and AI-generated code quality evaluation techniques	Aug 2025 – Feb 2026
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Perceived Stress Prediction Using Machine Learning with Feature Selection Python, XGBoost, Scikit-learn - Developed machine learning models to predict perceived stress levels using EMA stress and sleep datasets from the StudentLife research study - Applied feature engineering, mutual information-based feature selection, and cross-validation techniques using XGBoost, SVM, Random Forest, and Neural Networks	Sep 2025 – May 2026
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Flow Pilot: Enterprise AI Workflow Automation Platform Next.js, TypeScript, PostgreSQL - Built a full-stack enterprise AI workflow automation platform supporting workflow orchestration, approval systems, governance dashboards, and API-driven operations - Developed scalable backend workflows and database-integrated features using Next.js, TypeScript, PostgreSQL, Prisma, and modern full-stack engineering practices	Jan 2026 – May 2026
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TECHNICAL SKILLS

Languages:	Python, Java, SQL, JavaScript, TypeScript, C/C++
AI / Machine Learning:	Machine Learning, NLP, LLMs, Transformers, Scikit-learn, XGBoost, PyTorch, Machine Learning Pipelines
Generative AI:	Prompt Engineering, AI Agents, RAG Systems, Hugging Face, LLM Workflows
Frameworks / Tools:	AWS, Docker, Kubernetes, Git, Airflow, MLflow
Software Engineering:	REST APIs, Backend Systems, System Design, CI/CD, Agile, SDLC
Concepts:	Data Structures & Algorithms, OOP, Scalable Systems, Intelligent Automation

PUBLICATIONS

- Sekar, T., & ElKharboutly, R., “Assessing AI-Based Code Generation Using Code Comments,” 2026 5th International Conference on Information Systems Engineering (IISec), IEEE Xplore, 2026
- Sekar, T., ElKharboutly, R., & Sahoo, S., “Perceived Stress Prediction Using Machine Learning with Feature Selection on EMA Stress and Sleep Data: A StudentLife Study,” at IEEE/ACM CHASE 2026 Poster & Demo Track